Journal of Intelligent Information Systems (2024) 62:1659–1683
https://doi.org/10.1007/s10844-024-00855-y

RESEARCH

Aspect sentiment triplet extraction based on data augmentation and task feedback

Shu Liu[1] · Tingting Lu[1] · Kaiwen Li[1] · Weihua Liu[2]

Received: 17 November 2023 / Revised: 12 March 2024 / Accepted: 13 March 2024 /
Published online: 20 May 2024

Abstract
Aspect sentiment triplet extraction (ASTE), which focuses on mining the triplets (aspect, opinion, sentiment), is a complex and integrated subtask in aspect-based sentiment analysis. It is widely used in market research, product design and promotion, online comment analysis, and so on. Although significant progress has been achieved in existing methods, several challenges remain, such as data scarcity and the separation of span extraction and sentiment classification. Therefore, this paper adds data augmentation and task feedback based on the bidirectional machine reading comprehension model. Before training the model, the data augmentation module applies mask prediction and mark replacement to enrich the data. Span extraction and sentiment classification are two tasks during ASTE. We adopt the direct span extraction method with one classifier to avoid the error accumulation caused by multiple classifiers and to improve the adaptive ability between different datasets. In addition, we fuse the text features derived from the above two tasks for sentiment classification. Based on the feature fusion, the task feedback module is established to alleviate the task separation. Extensive experiments verify the effectiveness of our method. The code is available at https://github.com/zhenzhen313/BMRC-with-DA-and-TF.

Keywords Aspect sentiment triplet extraction · Bidirectional machine reading comprehension · Data augmentation · Task feedback · Feature fusion

✉ Weihua Liu
liuweihuayjy@chinamobile.com

Shu Liu
sliu35@csu.edu.cn

Tingting Lu
lutt0109@csu.edu.cn

Kaiwen Li
itkaven@csu.edu.cn

1 School of Computer Science and Engineering, Central South University, Changsha, China

2 China Mobile Research Institute, Beijing, China

1 Introduction

With the rapid growth of social media platforms, monitoring public opinion to facilitate decision-making become a constant demand. For example, understanding public sentiment on social events is vital for policy modification and law enforcement. Companies typically consider customer opinions and attitudes toward their products when determining the development direction in the next quarter. Therefore, aspect-based sentiment analysis (ABSA) has attracted widespread attention from businesses and researchers (Hu & Liu, 2004; Singh & Singh, 2021). ABSA is an important research field in natural language processing, which aims to mine the fine-grained opinions towards specific aspects and analyze the sentiment polarity. It involves three essential subtasks, i.e., aspect term extraction (ATE) (Toh, 2014), aspect-opinion pair extraction (AOPE) (Zhang & Liu, 2014), and aspect-level sentiment classification (ASC) (Lin & He, 2009; Zhu et al., 2023).

Aspect sentiment triplet extraction (ASTE) (Xu et al., 2020; Peng et al., 2020) is a more complex and integrated task in ABSA, requiring joint learning with multiple subtasks. It focuses on extracting the triplets (aspect, opinion, sentiment), for example in Fig. 1, where each contains an aspect expressing the sentiment, an opinion corresponding to the aspect, and a sentiment polarity associated with the aspect and the opinion.

Peng et al. (2020) initially introduced the ASTE task and proposed a two-stage pipeline method to deal with it. Specifically, two sequence labelings were performed to extract aspects, opinions, and corresponding sentiment polarities. The proper pairs of aspect and opinion were then classified from predicted terms, and finally construct the triplets. To take the advantage of the relationships among multiple sentiment elements, some studies have developed the unified methods. Zhang et al. (2020) proposed a multi-task learning framework, including ATE, opinion term extraction (Dave et al., 2003; Ravi & Ravi, 2015) and sentiment dependency parsing (Di Caro & Grella, 2013), where the triplets were generated from the predictions of these subtasks by heuristic rules. Another potential idea is to design a unified labeling method to extract the triples simultaneously (Xu et al., 2020; Wu et al., 2020). Xu et al. (2020) considered an extension of the unified labeling scheme (Li et al., 2019), namely position-aware tagging, and leveraged the location information of opinions. However, the above methods rely on the interaction between word pairs. In other words, the performance could be poor if the aspects or opinions are expressed in several words. Motivated by this observation, Xu et al. (2021) proposed a span-level interaction model that considered the entire span of interactions between aspects and opinions. Zhang et al. (2022) not only exploited the span-based methods to extract multiple entities but also considered the semantic relation between aspect

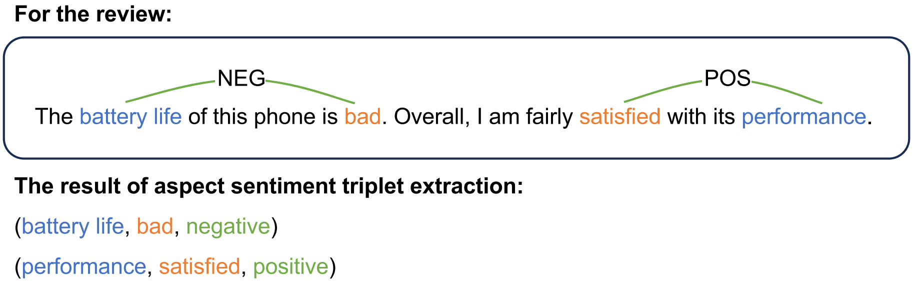

Fig. 1 An example of ASTE task. The aspects, opinion expressions, and sentiments are marked with blue, red and green, respectively

and opinion terms. Due to the excellent performance of graph convolutional network (GCN) in processing complex structural information, AlBadani et al. (2023) and Wang et al. (2023) apply it to handle correspondence between terms for ASTE.

There are other modeling paradigms to handle ASTE, such as machine reading comprehension (MRC) (Chen et al., 2021; Mao et al., 2021; Liu et al., 2022) and sequence to sequence (Seq2Seq) (Zhang et al., 2021; Yan et al., 2021). Mao et al. (2021) designed specific queries to transform the original extraction task into two MRC tasks, where the first for aspect extraction and the second for corresponding opinion and sentiment prediction. The bidirectional MRC (BMRC) framework in Chen et al. (2021) adopted a similar approach. One direction first predicted the aspects and then corresponding opinions, while the other direction sequentially detected the opinions and aspects described by opinion. Such bidirectional queries are expected to improve the model robustness. Furthermore, Seq2Seq modeling provides an elegant solution to identify the triplet elements at once. Zhang et al. (2021) recast the original task as a text generation problem and proposed two modeling paradigms for triplet prediction. Yan et al. (2021) took sentences as input, and output the sequences that consist of a mixture of term pointer indexes and sentiment class indexes.

Although significant progress for ASTE has been achieved in the past few years, several challenges remain. Typically, the advancement of deep learning is mainly promoted by large-scale labeled datasets. It is necessary to annotate the sentiment polarity of each aspect, which however, is time-consuming and labor-intensive. The scarcity of sufficient labeled data makes it difficult to develop a powerful learning model.

Span extraction and sentiment classification are primary tasks in ASTE. However, most previous studies have treated them as independent and ignored their complementary relationship, handling each separately. BMRC first extracts spans, followed by the sentiment classification task, resulting in the triplets. If extracted spans are inaccurate, resulting in suboptimal aspect-opinion pairs, then performing sentiment analysis on these incorrect pairs is highly problematic. Aspect-opinion pairs represent information in the text containing sentiment attributes, meaning spans and sentiments are highly correlated in some cases. Therefore, the loss of the sentiment classification task should be propagated to the span extraction task.

In this paper, we introduce data augmentation and task feedback into the backbone BMRC for ASTE. We first enrich the scale of the datasets through the data augmentation module, including mask prediction and mark replacement. Compared to the original BMRC, we advance the classifier for span extraction that can avoid accumulated errors and better adapt to different datasets without additional span matching. Then, the features derived from the span extraction and sentiment classification extractors are fused to optimize the performance of triplet recognition. The span and sentiment interaction can provide critical information for each other and further improve the model's performance. Due to the feature fusion, the task feedback module between the two tasks is established. Overall, our contributions can be summarized as follows:

- In order to mitigate the effect of data scarcity, we proportionally mix the data generated by mask prediction and mark replacement based on the original dataset for model training.
- We adopt the direct span extraction method with one classifier to adaptively learn different data distributions instead of multiple classifiers in the original BMRC which leads to error accumulation by separate predictions.
- We establish the task feedback module that introduces feature fusion, which can address the separation of span extraction and sentiment classification.
- Extensive experiments on benchmark datasets show the effectiveness of our proposed method.

2 Related work

2.1 Aspect sentiment triplet extraction

Compared to traditional single-task models in ABSA, ASTE models can provide more sentiment information in text, and the ASTE task was initially introduced by Peng et al. (2020).

To effectively utilize the relationships between various sentimental elements, Zhang et al. (2020) introduced a multi-task framework, enabling the joint prediction of aspects, opinions, and sentiment dependencies. Chen et al. (2020) developed a relation-aware framework that facilitates collaborative work among sub-tasks through multi-task learning and relationship propagation mechanisms in a stacked multi-layered network. Some researchers extract triplets using unified tagging schemes such as the position-aware tagging scheme (Xu et al., 2020), grid tagging scheme (Wu et al., 2020), and span-level interaction models (Xu et al., 2021). Besides, Seq2seq can be used to predict triplets in one shot (Zhang et al., 2021; Yan et al., 2021; Fei et al., 2023). Fei et al. (2023) modeled the ASTE task as an unordered triplet set prediction problem and presented a non-autoregressive decoding method.

Recently, some studies have introduced new methods to improve the ASTE task. Chen et al. (2022a) predefined ten types of relationships and employed an enhanced multi-channel graph convolutional network model to utilize relationships between words. Chen et al. (2022b) proposed aspect and opinion decoders to decode span representations and extract triplets from aspect-to-opinion and opinion-to-aspect directions, thus achieving a more comprehensive extraction of triplets. Zhou et al. (2023) introduced two bidirectional templates for filter false paired triplets and a maker-oriented sequence labeling module to enable better handling of complex structures, overcoming the limitations of token-by-token decoding and simple structured prompt in traditional generative models. Yu et al. (2023) designed a retriever feting semantically and sentimentally similar triplets. The ASTE framework is trained jointly with the retriever to overcome the special challenges of retrieval-based methods.

When handling ASTE, most studies considered span extraction and sentiment classification as separate tasks without connection. Due to the lack of span information, sentiment classification may be limited, but the impact of spans on sentiments is hard to capture. In some cases, spans and sentiments are highly correlated. Additionally, for certain aspects, there may be multiple sentiment polarities with different opinions. Therefore, attention to the connection between span extraction and sentiment classification can affect the effectiveness of ASTE.

2.2 Data augmentation

Data augmentation is a widely employed preprocessing technique in deep learning. It refers to strategies that increase the quantity and diversity of training data, thus mitigating overfitting. Due to the common challenge of data scarcity in deep learning methods, it has garnered significant attention. In ASTE, data scarcity is also a common problem.

Li et al. (2017) enhanced the model's understanding of syntax and linguistic structure by generating noise with syntactic and semantic methods and word dropout. In order to improve the model's semantic understanding capabilities, methods such as synonym replacement (Wei & Zou, 2019) and mixed-based approaches (Sun et al., 2020) have emerged. AEDA (Karimi et al., 2021), randomly inserting punctuation marks into the original text, seeks to enhance the model's robustness and resilience and address diverse irregular language phenomena that may arise in real-world language data. Text AutoAugment (Ren et al., 2021) proposed a compositional and learnable paradigm to search for the optimal compositional policy automatically and improve the diversity and quality of augmented data. The studies above have achieved competitive performance on conventional classification tasks, but few works are applied to ABSA.

In ABSA, researchers have also developed various data augmentation methods. Wang et al. (2021) propose a progressive self-training framework on the unlabeled data. Liang et al. (2021) deployed a supervised contrastive learning framework with aspect-invariant/-dependent data augmentation. Wang et al. (2022) augmented both aspects and polarities. However, most existing studies tend to either modify the labels of the given instances, require the collection of more data, or employ complex methodologies. Our approach is uniquely designed for ASTE, focusing on enhancing performance without requiring additional data. We maintain the labels of the original instances, aiming to develop concise and practical data augmentation techniques.

3 Methodology

3.1 Overview

In this work, BMRC (Chen et al., 2021) is utilized as the backbone network. It formalizes ASTE as a problem of machine reading comprehension, which facilitates query creation based on context and span extraction based on the answer. By constructing bidirectional queries, BMRC improves the performance of span extraction to advance the final sentiment classification. Here we divide all queries in BMRC into two tasks, span extraction and sentiment classification.

As shown in Fig. 2, two modules are added based on the backbone. The data augmentation module plays the role of enriching the training data. The task feedback module deals with the separation of two tasks in the original BMRC. The dotted line represents the relationship between span extraction and sentiment classification, which forms a feedback structure.

3.2 BMRC

The goal of machine reading comprehension is to enable a model to understand and respond to questions about a given text like human comprehension. BMRC utilizes complementary information from both directions($A \rightarrow O$ and $O \rightarrow A$), facilitating the recognition of more comprehensive triplets.

$A \rightarrow O$ All aspect terms are retrieved with $Query\ A$ from the text based on the context, and the corresponding opinion term set is queried with $Query\ \ A \rightarrow O$ based on those aspects.

$O \rightarrow A$ All opinion terms are retrieved with $Query\ O$ from the text based on the context, and the corresponding opinion term set is queried with $Query\ \ O \rightarrow A$ based on those opinions.

The aspect-opinion pairs can be obtained by merging the answer sets of $Query\ A \rightarrow O$ and $Query\ \ O \rightarrow A$. This information can be used to construct $Query\ S$ that predicts corresponding sentiments based on the context. Finally, combining aspect-opinion pairs with predicted sentiments results in aspect-sentiment triplets. A specific example is shown in Fig. 3.

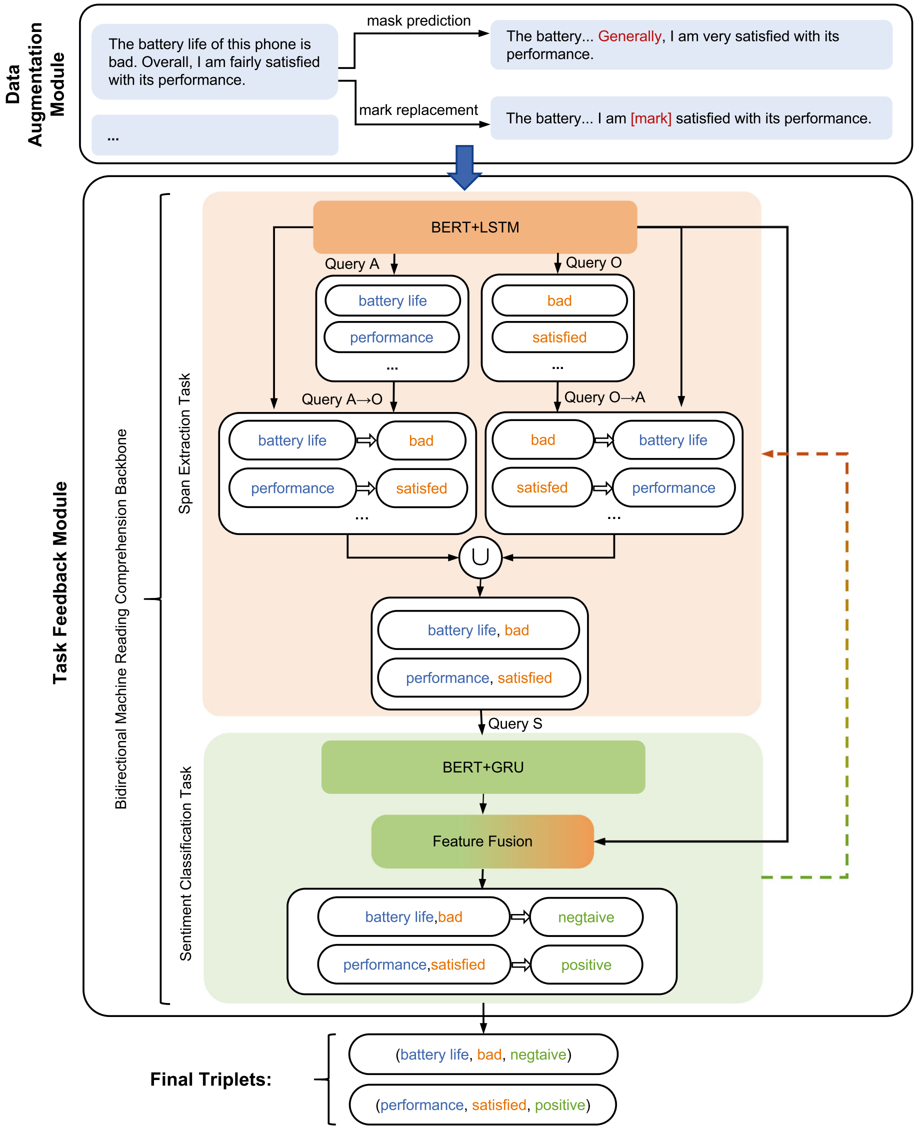

Fig. 2 The overview of our method. Data augmentation is introduced to generate a new dataset to be fed into BMRC. Aspects and opinions are captured during the span extraction task. Next, the corresponding sentiments are classified during the sentiment classification task. Task feedback is established between the two tasks to connect them. Finally, aspect sentiment triplets can be obtained

3.3 Data augmentation

Data augmentation is extremely valuable in ASTE. Since ASTE often encounters data scarcity, data augmentation can use various techniques to generate new training data, leading to notable improvements in the model's generalization ability and overall performance. Firstly, data augmentation can broaden the scope of available data and introduce diverse

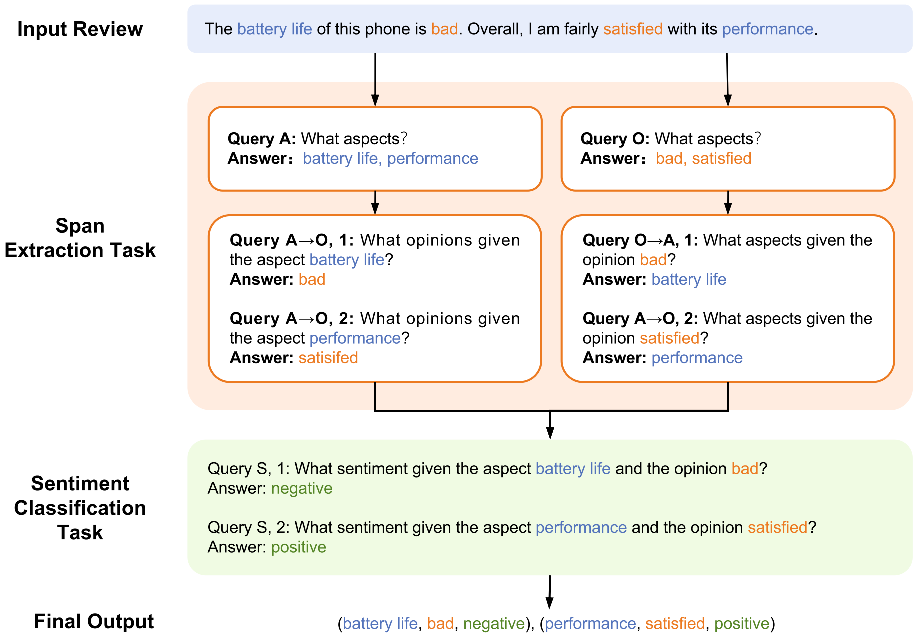

Fig. 3 The structure of BMRC

scenarios absent in the original dataset. It improves the model's performance within the confines of the training data and equips it to excel in novel situations, enhancing its adaptability and robustness. Furthermore, data augmentation is vital in mitigating the risk of overfitting. ASTE often involves challenges such as small datasets, multiple labels, and high correlations among data. Data augmentation addresses this by increasing the training data volume while preserving the integrity of the original data distribution. Additionally, diversifying the dataset makes the model more tolerant to noise, mislabeling, and other problems, fostering the model to handle real-world challenges more effectively.

Given the complexity and integrity of ASTE, we have a higher standard for the quality of the samples. Through data augmentation, the training data should be more extensive and diverse so that the model can be designed to be more complex. Moreover, the labels should not be changed to ensure the reliability of the extracted triplets as much as possible. We aim to avoid the time-consuming collection of additional data or complex data processing. Therefore, we design several simple but effective strategies to optimize the generalization ability of the ASTE model.

3.3.1 Mask prediction

In the pre-training process in the BERT model (Devlin et al., 2019), some tokens were masked in the sequence at random and then the model predicted them to learn semantic relationships and contextual information. Inspired by this, we replace some words with a particular tag [MASK] and then use BERT to recognize these tags based on context, so that the recognized results are usually close to or related to the original words. Finally, we generate new texts from these recognitions. Since the masked positions are random, the data diversity can be expanded to alleviate the overfitting. An example of mask prediction is illustrated in Fig. 4.

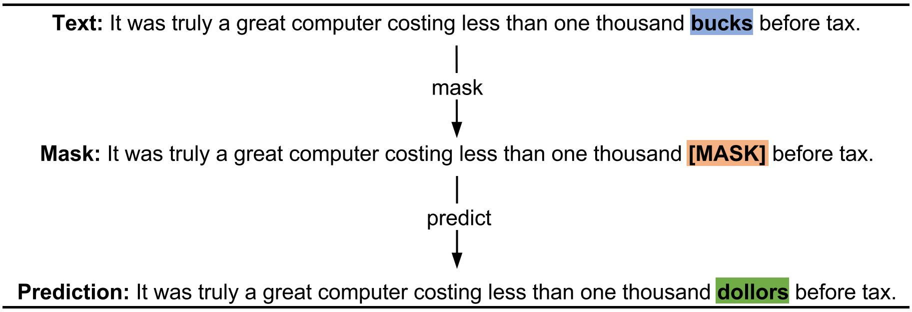

Fig. 4 Illustration of mask prediction. The word **bucks** in the original text is tagged with **[MASK]**. Then we use BERT to predict the tag and replace it with the predicted result **dollars** to get the new text

3.3.2 Mark replacement

In BERT, the token [UNK] represented unknown words or sequences, which were learned to be the appropriate words during training. We also replace some words in the text with such tokens (Fig. 5), enabling the model to infer unknowns based on contextual information and deal with them more effectively. However, if the token [UNK] is used excessively, the model may misinterpret the text and behave poorly. Therefore, the proportion of such tokens in the new text needs to be reasonable, and the same is true for such new texts in the training data.

3.3.3 Proportional mixing

Based on the original dataset, we generate the new data using the above two strategies and then mix them proportionally. Both the original and the new texts are used for model training. Note that we set a constraint that the number of masked or replaced words is less than 20% of the total words in the text, thus the potential uncertainty caused by excessive word masking or replacement could be prevented. As for proportional and convenient mixing, these words to be augmented are masked first. Aspect and opinion terms are not within the masked area, so there is no need to change the original labels. Then one random integer from 1 to 100 is generated to regulate the proportion of data obtained from both augmentation methods. If the integer is greater than 20, the masked words in the text are predicted; otherwise, the [MASK] tags are replaced with [UNK]. In this sense, the data proportions via mask prediction and

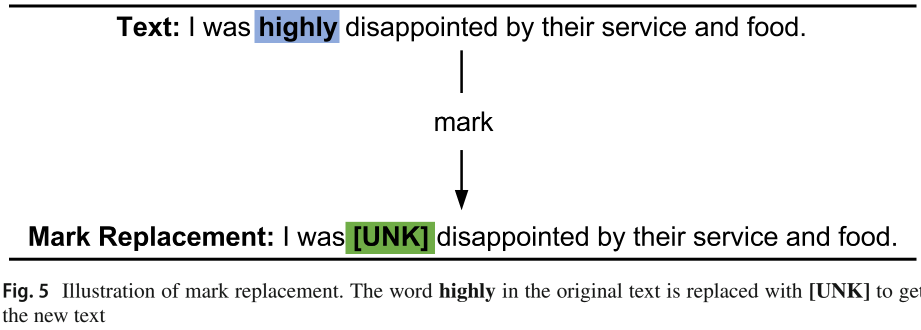

Fig. 5 Illustration of mark replacement. The word **highly** in the original text is replaced with **[UNK]** to get the new text

mark replacement are around 80% and 20%, respectively. We also conduct some experiments to verify the proportion of the generated data in Section 4.3.

3.4 Span extraction

The span extraction task focuses on extracting term spans of aspects and opinions. The sentence X with N tokens will be concatenated with the corresponding queries format as $\{[CLS], q_1, q_2, ..., q_{|q|}, [SEP], x_1, x_2, ..., x_N\}$ to gain the corresponding spans, where $[CLS]$ is the beginning token and $[SEP]$ is the segment token. First, We input the sentence X to the BERT model to obtain the hidden representation sequences $H = \{h_1, h_2, ..., h_{|q|+2+N}\}$.

To fully learn the semantic dependence of input sequences, we feed the hidden representation H to LSTM to obtain the feature representation of the entire sequence.

$$F^{span} = \{f_1^{span}, f_2^{span}, ...f_{|q|+2+N}^{span}\} = LSTM(H, \theta_l) \tag{1}$$

where θ_l is the parameter for the LSTM. Therefore, our $SpanExtractor$ is constructed by the above two models. Then, a multiclass classification is used to determine whether the tokens belong to the spans, followed by a softmax function to compute the probability distribution of the span prediction for each token. An example of span extraction is illustrated in Fig. 6.

Since multiple classifiers are used for span extraction in the original BMRC, there is a risk of error accumulation caused by separate predictions of the start and end positions. Instead, we adopt a direct span extraction method with one classifier, described by

$$\hat{y} = f(X) = [\hat{y_1}, \hat{y_2}, ..., \hat{y_{n-1}}, \hat{y_n}] \tag{2}$$

where $\hat{y_i}$ represents whether the i-th token is a component of the span. The value of $\hat{y_i}$ could be: 0 – neither the start nor the end position of a span, 1 – the start position of a span, 2 – the end position of a span, 3 – both the start and the end position of a span (or the token is a span). Specifically, we predict whether the token x_i is the span of the answer or not, followed as:

$$p(\hat{y_i}|x_i, q) = Softmax(f_{|q|+2+i}^{span} W_{span}) \tag{3}$$

where $W_{span} \in \mathbf{R}^{d_h \times 4}$ is the model parameter, d_h denotes the dimension of feature representations in F^{span}, and $|q|$ is the query length.

In addition, two binary classifiers are used in BMRC to predict whether each position is the start or end of a span separately. The predictions are then converted to probabilities using the softmax function. Since the predicted start and end positions are separate, it is essential to match them to identify aspects or opinions. Compared to BMRC, the direct span extraction method is widely applicable. This method can predict all the spans in the text at once without any additional span matching, and adjust the classifier parameters according to different datasets. Notably, this process only matches normal outputs: (1) The position of Value 2 is behind Value 1. If not, the model will ignore these values. Value 2 only matches the front Value 1. (2) Value 3 is outside the range of Value 1 and Value 2. Except for Value 0, the closest value on the left can only be Value 2, and the closest value on the right can only be Value 1.

We use the span extraction strategy mentioned above to obtain the probabilities for each aspect and opinion's corresponding start and end positions without requiring matching. Since

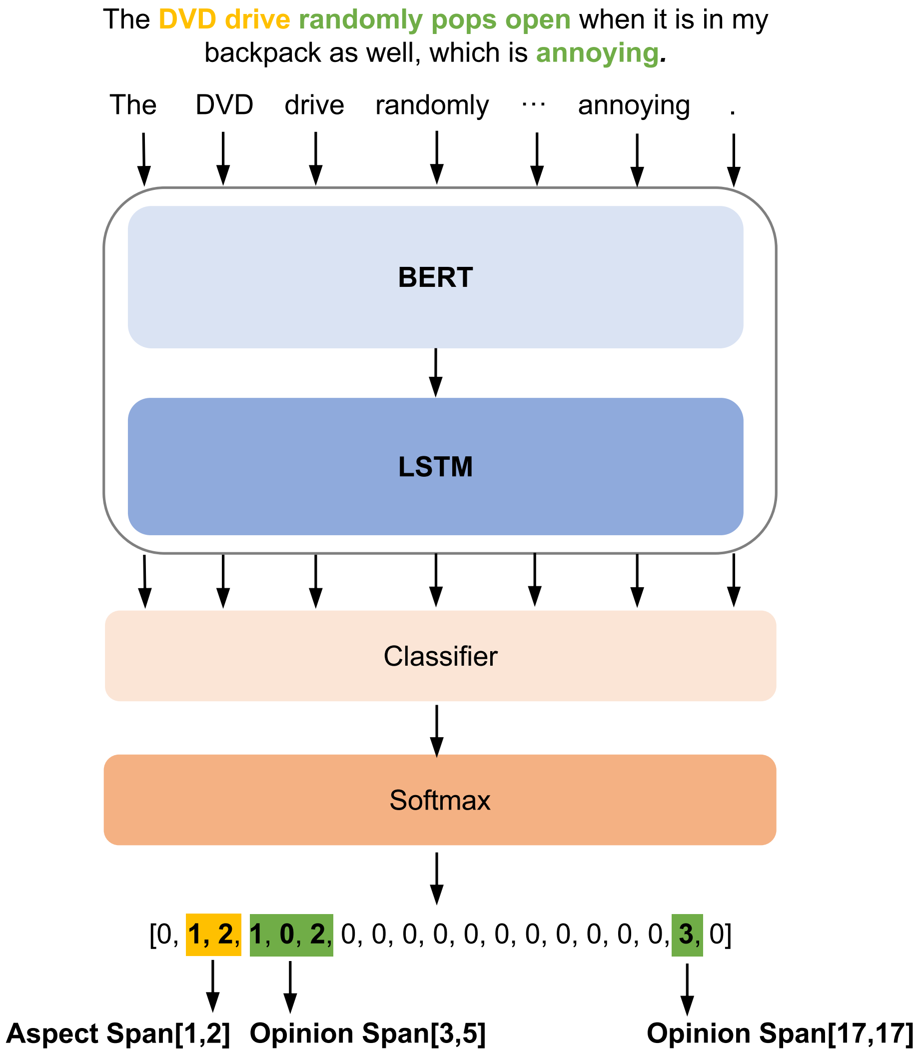

Fig. 6 Illustration of span extraction. The sequence at the bottom of the figure is the output of the span extraction model. The spans in yellow represent the aspect term **DVD drive**. The spans in green represent the opinion terms **randomly pops open** and **annoying**, respectively

the start and the end of aspects and corresponding opinions span either one or two positions, their predicted probabilities are calculated by Eqs. 4 and 5, respectively.

$$Aspect_{probability} = \sqrt{Aspect^{start}_{probability} * Aspect^{end}_{probability}} \tag{4}$$

$$Opinion_{probability} = \sqrt{Opinion^{start}_{probability} * Opinion^{end}_{probability}} \tag{5}$$

where $Aspect^{start}_{probability}$, $Aspect^{end}_{probability}$, $Opinion^{start}_{probability}$ and $Opinion^{end}_{probability}$ individually represent the predicted probability of the start and end location of the aspect and opinion in the text.

BMRC relies on the product of the start and end positions' probabilities to determine the probability of a span. The product of the probabilities of aspect and opinion determines the probability of the aspect-opinion pair. This approach decreases the pairing probabilities and may not accurately reflect the aspect-opinion pair prediction. For example, for an aspect-opinion pair, if both the start and end positions of the aspect and opinion have a probability of 0.8, then the probability of the aspect-opinion pair will be $0.8^4 = 0.4096$. Our calculations could smooth the probabilities and advance the model more robust. The probabilities of aspects and opinions fall within the range of predicted probability values that encompass start and end positions, rather than simply performing multiplication. In other words, the decrease in predicted probability values can be avoided. Equation 6 defines the calculation of the predicted probability of the aspect-opinion pair. If the pair is extracted in both $A \to O$ and $O \to A$ directions, it will be considered as the predicted aspect-opinion pair. Otherwise, if the probability of the pair is higher than the given threshold δ, it will also be included in the final results.

$$Pair_{probability} = \sqrt{Aspect_{probability} * Opinion_{probability}} \tag{6}$$

3.5 Sentiment classification

The sentiment classification task aims to classify sentiment polarities, including positive, negative, and neutral. Our model uses BERT and GRU for feature extraction and then fuse the features derived from span extraction and sentiment classification. Finally, we use the three-class classification task for sentiment polarity prediction. The specific model structure is shown in Fig. 7.

GRU and LSTM are common recurrent neural network structures for sequential data processing. GRU is a more simplified model that combines certain aspects of the LSTM architecture. GRU does not have a separate memory unit and directly uses the hidden state to transfer information. This makes GRU generally more efficient computationally but potentially less effective at capturing complex long-term dependencies in data (Cahuantzi et al., 2023). Therefore, LSTM can better handle span extraction that requires memorizing long-term dependencies. GRU can perform better for simple sequential tasks such as sentiment classification. We input the last hidden state representation H of the BERT into the GRU model, obtaining the feature representation for the entire text sequence.

$$F^{sentiment} = GRU(H, \theta_g) \tag{7}$$

where θ_g is the parameter for the GRU. Therefore, Our $SentimentExtractor$ is constructed from the BERT and GRU models.

Feature fusion combines different feature representations to create a more comprehensive and integrated representation. In deep learning, if the features from different layers or neural networks capture diverse data, combining them through feature fusion can gain a more comprehensive representation to make decisions.

We propose a feature fusion method that fuses span-level features with sentiment-level features to get the final feature representation for sentiment classification task, followed as:

$$F^{fusion} = \mathcal{C}((1-\alpha) * F^{span}, \alpha * F^{sentiment}) \tag{8}$$

where $\mathcal{C}(\cdot)$ refers to concatenation in the second dimension. F^{span} is extracted by $SpanExtractor$, which is dedicated to capturing span information. $F^{sentiment}$ is extracted by $SentimentExtractor$, which pays attention to identifying sentiment information. The

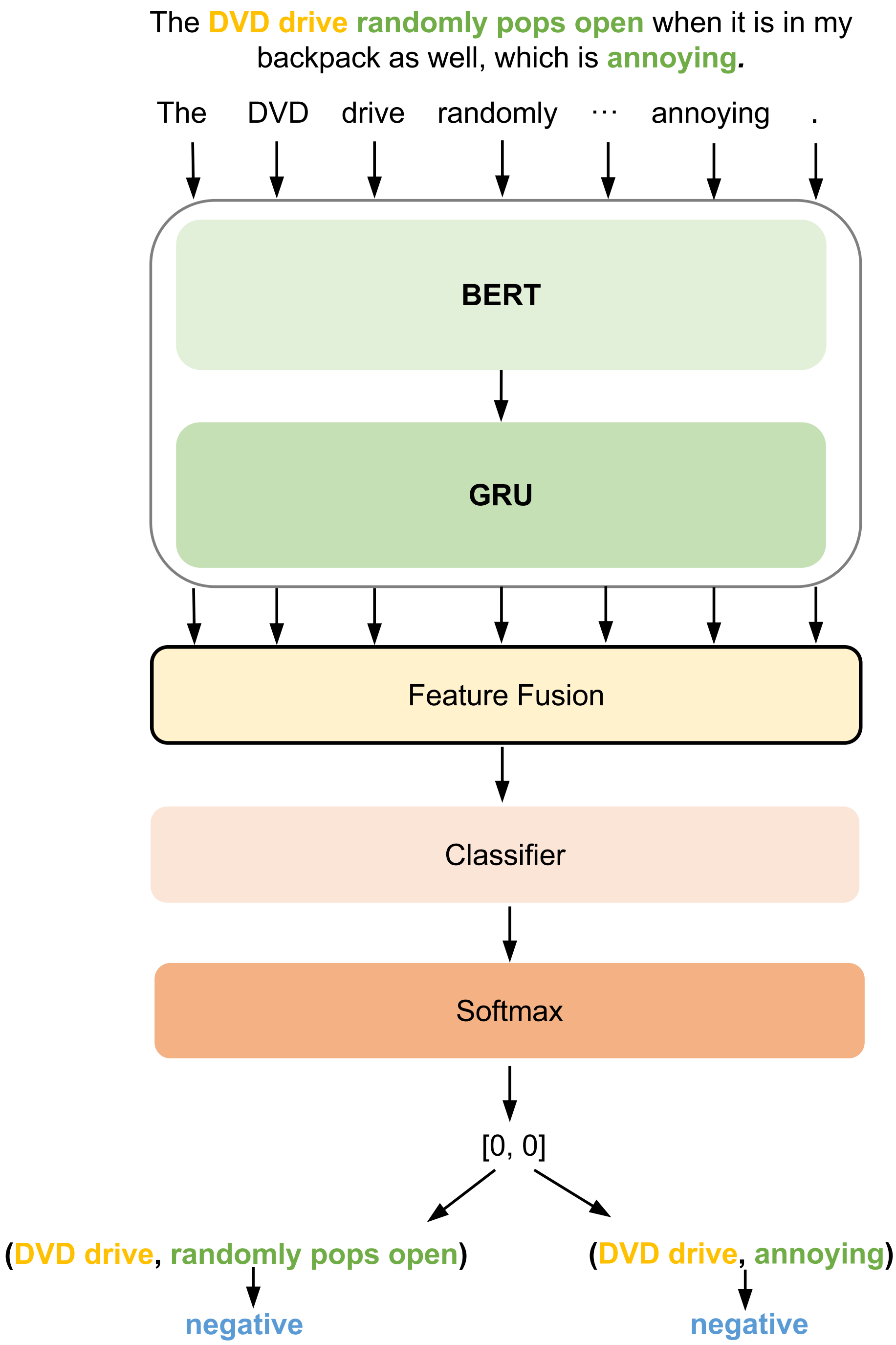

Fig. 7 Illustration of sentiment classification. The sentiment polarities at the bottom of the figure is the out of the sentiment classification model. The left **negative** represent the sentiment according to the aspect **DVD drive** and opinion **randomly pops open**. The right **negative** represent the sentiment according to the aspect **DVD drive** and opinion **annoying**

weight parameter α is used to control the contribution of those two extractors. Considering that the sentiment classification task is expected to capture more information containing sentiments in the text, we set α to 0.8 in our experiments. An ablation study of different α will be shown in Section 4.3 to validate this setting. Finally, we use a three-class classifier to predict the sentiment polarity $\hat{y}^{\mathcal{S}}$ as follows:

$$p\left(\hat{y}^{\mathcal{S}} \mid X, q\right) = Softmax\left(F^{fusion} W_{sentiment}\right) \tag{9}$$

where $W_{sentiment} \in \mathbf{R}^{d_f \times 3}$ is the model parameter, d_f denotes the dimension of the fused feature representations in F^{fusion}.

3.6 Task feedback

Tasks of span extraction and sentiment classification are inherently complementary. The performance of sentiment classification depends on the information derived from span extraction. On the other hand, the results of sentiment classification can be used to guide span extraction. However, most studies ignore the relationship between these two tasks. The multi-information interaction can provide important information for each other and further improve the performance of the model. It is essential to establish a connection between span extraction and sentiment classification.

To this end, features derived from both tasks are fused for sentiment classification as mentioned in Section 3.5. The fused features contain global sentiment features and span-related semantic information. As shown in Fig. 8, for the span extraction task, we obtain the span features by $spanExtractor$ and then use them to predict the term spans. For the sentiment classification task, we input the fused feature to the classifier, and calculate the

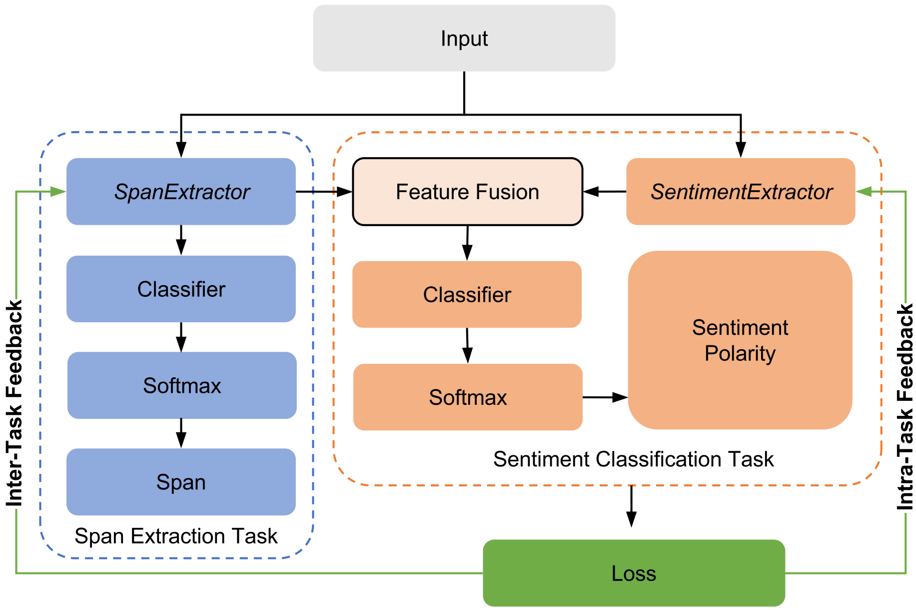

Fig. 8 Illustration of task feedback. Due to feature fusion, the loss of sentiment classification is feedback to span extraction and itself

output by the softmax function to obtain the sentiment polarity prediction. The cross-entropy loss is always used as the loss function in sentiment classification. The model parameters are optimized to align their predicted labels with the ground truth by minimizing the loss of sentiment classification as follows:

$$\mathcal{L}_{\mathcal{S}} = -\sum_{i=1}^{\left|Q^{\mathcal{S}}\right|} p\left(y^{\mathcal{S}} \mid X, q_i^{\mathcal{S}}\right) \log \hat{p}\left(\hat{y}^{\mathcal{S}} \mid X, q_i^{\mathcal{S}}\right). \tag{10}$$

where $p(*)$ represents the gold distribution, $\hat{p}(*)$ denotes the predicted distribution, $\left|Q^{\mathcal{S}}\right|$ is the number of the sentiment queries.

Because of feature fusion, the loss of sentiment classification is not only propagated back to $SentimentExtractor$, but also to $SpanExtractor$ to improve their performance. Especially, if the extracted spans are incorrect, the classified sentiments may not be as expected. Therefore, it is essential to feed back the loss to $SpanExtractor$ so that it can adapt to the current sentiment classification task. The propagation to $SentimentExtractor$ is intra-task feedback, and the propagation to $SpanExtractor$ is inter-task feedback. Based on those feedback, our method effectively avoids issues caused by the separation of the two tasks.

Hence, once the loss of sentiment classification exists, it is fed back to the span extraction task, facilitating the two tasks to pay more attention to others' information. The spans in the text carry sentiment information, and the sentiments are the representation of span information. In this sense, the model performance is finally advanced by the task feedback module.

4 Experiments

In this section, extensive experiments are conducted to verify the effectiveness of our method. We first briefly introduce the experimental setup (Section 4.1). Then, we discuss the effectiveness of each module, evaluate the the proportion of mask prediction and mark replacement and α (Section 4.2), and we compare our method with state-of-the-art methods (Section 4.4). Finally, compare the performance in several subtasks (Section 4.5).

4.1 Experimental setup

4.1.1 Datasets

We evaluate the proposed method on popular datasets ASTE-Data-V2-EMNLP including 14res, 14lap, 15res, and 16res released by Xu et al. (2020) which refined the missing triplets that are not explicitly annotated in the previous version (Peng et al., 2020). All of them are derived from benchmark datasets on the SemEval ABSA challenge. Both l4res and 14lap are constructed from SemEval 2014 Task 4 (Pontiki et al., 2014), 15res from SemEval 2015 Task 12 (Pontiki et al., 2015) and 16res from SemEval 2016 Task 5 (Pontiki et al., 2016). The aspects, opinions, and their sentiment polarities are labeled. Table 1 lists the statistical data of these benchmarks published by SemEval from 2014 to 2016, regarding laptops, cameras, restaurants, and hotel reviews.

Table 1 Statistical data of four benchmark datasets, #S and #T denote the number of sentences and triplets, respectively. #+, #0, and #- denote the numbers of positive, neutral, and negative triplets respectively

Datasets	Train					Dev					Test				
	#S	#T	#+	#0	#-	#S	#T	#+	#0	#-	#S	#T	#+	#0	#-
14res	1266	2338	1692	166	480	310	577	404	54	119	492	994	773	66	155
14lap	906	1460	817	126	517	219	346	169	36	141	328	543	364	63	116
15res	605	1013	783	25	205	148	249	185	11	53	322	485	317	25	143
16res	857	1394	1015	50	329	210	339	252	11	76	326	514	407	29	78

4.1.2 Performance metrics

For evaluating the model performance, we take the F1 score as the main performance metrics for all experiments. In addition, the evaluation metrics in Section 4.4 also include precision (P), and recall (R). Only if the aspect, opinion, and sentiment are correct, the predicted triplet will be correct.

4.1.3 Implementation details

For the encoding layer, we adopt the BERT-base-uncased model with a hidden size of 768. Our proposed method is implemented with the PyTorch toolbox on an RTX 3090 GPU. During training, we use AdamW (Loshchilov & Hutter, 2017) optimizer with the initial learning rate of 2×10^{-4}. The fine-tuning rate for BERT is 2×10^{-5}. The total number of training epochs and the batch size of data are set to 30 and 8, respectively. These setups are the same for all the datasets. The threshold δ is manually tuned in bound [0, 1) according to the triplet extraction F1 score on the development sets, and we set δ to 0.98, 0.98, 0.9 and 0.98 for 14res, 14lap, 15res and 16res datasets, separately.

4.2 Effectiveness of each module

To evaluate the importance of the proposed modules, we first utilize the BMRC (Chen et al., 2021) model without these modules to tackle the ASTE task. The BMRC model used in the experiment is a reproduction version of the initial model on the ASTE-Data-V2-EMNLP2020 datasets. Then, we conduct the experiments by introducing the data augmentation module and the task feedback module separately. Finally, we integrate the data augmentation and task feedback modules for ablation studies.

As shown in Table 2, several observations can be summarized as follows. The model with the data augmentation module experienced a more significant improvement in recall, leading to an overall rise in F1 score. Data augmentation increases the diversity of sentences and words in the training set, whether it involves noise, different types of data, or the same type of derived data. The model learns more information, allowing it to analyze more semantics and discern more target spans when faced with the same test set.

However, the model with the data augmentation module decreases the precisions when processing each dataset compared with the BMRC model. We can find that more aspect sentiment triplets are predicted, but a large part of them fail to match the target triplets in Fig. 9. Noise in the generated data can indeed mislead the model. These datasets primarily consist of information related to reviews of laptops, hotels, and other topics. If irrelevant reviews are

Table 2 Ablation studies of the data augmentation and task feedback modules on the test set of ASTE-Data-V2-EMNLP2020 datasets

Models	14res			14lap			15res			16res		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
	73.42	66.70	69.89	**69.53**	52.11	59.57	**64.06**	55.87	59.69	**70.37**	66.53	68.39
+ Data Augmentation	72.75	70.92	71.82	59.06	**62.98**	60.96	60.84	65.36	63.02	64.90	76.65	70.29
+ Task Feedback	70.75	71.53	71.14	59.93	62.80	61.33	60.89	65.15	62.95	68.21	74.31	71.14
Ours	73.06	**74.24**	**73.65**	66.26	60.03	**62.99**	63.79	**67.21**	**65.46**	68.97	**77.43**	**72.96**

The bold emphasis reflected the best metric performance

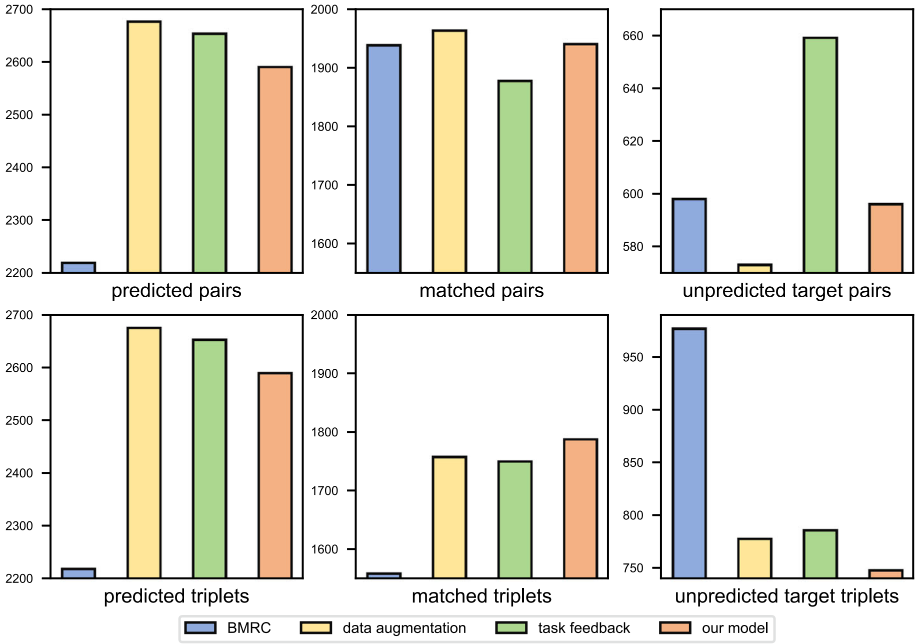

Fig. 9 Visualization of the number of predicted, matched and unpredicted target pairs and triplets extracted by different models

generated, it may introduce interference to the model. Furthermore, the data augmentation is randomly generated based on the training set, which may differ in distribution from the test set.

Task feedback creates a connection between span extraction and sentiment classification, enhancing the model's ability to discern sentiment triplets. This improved sensitivity not only enables more correct capture of emotional content in texts but also aids in identifying sentiment polarities. As a result, the model is more likely to form additional triplets, including those that may not directly align with the targets. Consequently, while there is a decrease in precision compared with BMRC, the significant increase in recall underlines the effectiveness of task feedback. This is particularly evident in the improvement of the overall metric, F1 score, which balances precision and recall, indicating a more comprehensive understanding of aspect sentiment triplets in texts.

Through data augmentation and task feedback, our model demonstrates excellent performance in F1 score for the ASTE task, which consists of multiple subtasks. Noise and diverse data reduce the model's precision and impact individual subtasks. Therefore, the impact of subtasks on triplet extraction will accumulate. For instance, if these factors cause the effect of two subtasks to become 90% of the original, the effect of the whole task will be about 81% ($0.90 * 0.90$). Figure 9 shows that more expected spans are extracted, but the expected triplets are slightly less. By establishing the connection between these tasks, they can provide mutual information when encountering anomalous data. Spans are the carriers of sentiments, $SentimentExtrator$ can learn more information related to span. Sentiments are the abstract representation of spans, and the $SpanExtractor$ learns more sentiment information, allowing for more precise identification of tokens related to sentiment. The effect is potentially 85.56% ($0.93 * 0.92$). In addition, the task feedback module also introduces a little decrease

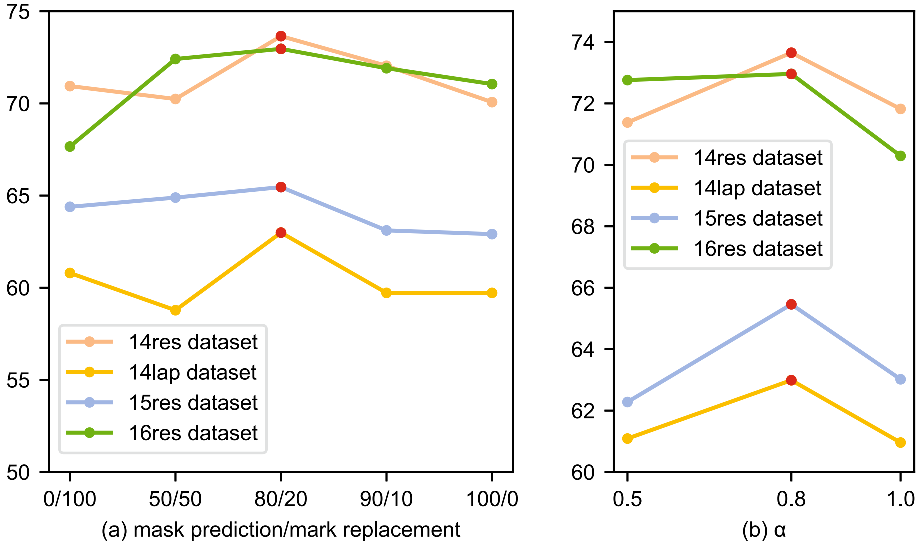

Fig. 10 **a** Performance for different proportions of mask prediction and make replacement in F1 score on the test set. **b** Performance for different parameter α in F1 score on the test set. The performance with our setting is marked in red

in precision. Therefore, our model in precision is higher than data augmentation alone, but it is still lower than the original BMRC model ($0.8556 < 1$).

4.3 Evaluation of parameters

Mask prediction and mark replacement are proposed to enrich datasets. The generated data will be mixed proportionally to the model. The performance under different proportion settings is plotted in Fig. 10(a). The model achieves optimal results on overall datasets when the proportion of mask prediction is set to 0.8, and the proportion of mark replacement is set to 0.2.

The parameter α is used for feature fusion to control the contribution of span extraction and sentiment classification for the final sentiment recognition. Since $SentimentExtractor$ can capture more sentiment information, the weight should be higher than $SpanExtractor$. Moreover, we can not obtain enough span information for feature fusion and task feedback if α is too large. Figure 10(b) compares the results of different $\alpha \in \{0.5, 0.8, 1\}$. Our method obtains the top performance when α is set to 0.8.

4.4 Comparison with state-of-the-art methods

The experimental results are presented in Table 3. Based on the strong baseline Span-ASTE (Xu et al., 2021) with excellent performance on spans containing several words, our method improves the F1 scores, i.e., 1.80% on the 14res, 3.61% on the 14lap, 2.19% on the 15res, and 2.70% on the 16res. The remarkable results demonstrate the effectiveness of our introduced data augmentation and task feedback.

Table 3 Comparison with state-of-the-art methods on the test set of ASTE-Data-V2-EMNLP2020 datasets

Models	14res			14lap			15res			16res		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
JET-BERT (Xu et al., 2020)	70.56	55.94	62.40	55.39	47.33	51.04	64.45	51.96	57.53	70.42	58.37	63.83
Span-BART (Yan et al., 2021)	65.52	64.99	65.25	61.41	56.19	58.69	59.14	59.38	59.26	66.60	68.68	67.62
GTS-BERT (Wu et al., 2020)	67.76	67.29	67.50	57.82	51.32	54.36	62.59	57.94	60.15	66.08	69.91	67.93
ETSR-BERT (Zhang et al., 2022)	70.72	68.86	69.79	58.63	54.95	56.75	60.58	63.86	61.70	68.32	70.23	69.26
EMC-GCN (Chen et al., 2022a)	71.21	72.39	71.78	61.70	56.26	58.81	61.54	62.47	61.93	65.62	71.30	68.33
Span-ASTE (Xu et al., 2021)	72.89	70.89	71.85	63.44	55.84	59.38	62.18	64.45	63.27	69.45	71.17	70.26
Chen et al. (2022b)	76.36	72.43	**74.34**	65.68	59.88	62.65	**69.93**	60.41	64.82	**71.59**	72.57	72.08
RIL (Yu et al., 2023)	**77.46**	71.97	**74.34**	63.32	57.43	60.96	60.08	70.66	65.41	70.50	74.28	72.34
RoBMRC (Liu et al., 2022)	72.51	72.73	72.62	**68.13**	57.09	62.12	65.90	65.36	**65.63**	69.98	76.65	**73.16**
Ours	73.06	**74.24**	73.65	66.26	**60.03**	**62.99**	63.79	**67.21**	65.46	68.97	**77.43**	72.96

The bold emphasis reflected the best metric performance

Compared to RIL (Yu et al., 2023) with a retriever to capture similar triplets and the method extracted triplet from both directions proposed by Chen et al. (2022b), our approach achieves a higher F1 score on the 14lap, 15res, and 16res datasets. However, on 14res, they demonstrate higher precision, contributing to an increase in F1 score compared to us. Chen's method exhibits good precision on most datasets, which may be a critical factor in achieving excellent performance in F1 score. Despite a slight decrease in precision, our method always performs well in recall, contributing to our excellent F1 performance on most datasets. Additionally, (Chen et al., 2022b; Liu et al., 2022) and our method extracted triplets from both directions and achieved better results, affirming the effectiveness of the bidirectional triplet extraction framework.

The method of this paper in F1 score impressively surpasses our previous work RoBMRC (Liu et al., 2022) by 1.03% and 0.87% on the 14 res and 14lap datasets. However, the F1 score of this paper are decreased by 0.17% and 0.20% on the 15res and 16res datasets respectively. The above results indicate that these two methods have their own advantages and disadvantages.

RoBMRC is a robustly optimized BMRC version that incorporates several improvements. It uses word segmentation for semantic learning and exclusive classifiers to avoid interference between different queries. In addition, it proposes a span matching rule and a probability generation strategy. Therefore, it can handle rare words with word segmentation, i.e., when words not present in the encoded vocabulary occur. Although the span matching proposed in RoBMRC can extract spans that better align with expectations, it is not applicable to adaptively learn feature distributions with significant differences between the data. Due to limitations in the data scale, the RoBMRC model cannot have a complex structure, which makes it more prone to overfitting. Compared to the method in this paper, the RoBMRC model is more straightforward to implement and requires fewer computational resources.

This paper focuses on introducing two modules on the backbone BMRC. Incorporating data augmentation makes it more suitable for uneven data distribution, such as significant differences in the number of various aspect terms in the dataset. Moreover, the design of the model structure is more complex than RoBMRC, which can reduce overfitting. However, it is essential to note that data augmentation may introduce noise that can significantly impact the performance of the model. The direct span prediction method is better suited for datasets with different feature distributions. It does not have the limitations of specific strategies, allowing it to learn different feature distributions adaptively. Furthermore, the performance of the task feedback module depends on two feature extractors. If the design of both feature extractors is inappropriate, it can lead to feature conflicts.

4.5 Comparison in multiple subtasks

To the best of our knowledge, ASTE is a challenging task that involves not only the extraction of aspects and opinions in the text but also the determination of their corresponding relationships and the sentiment polarities expressed. Due to its complexity, ASTE holds great practical value in real-world applications. In general, ASTE typically involves the collaborative completion of multiple subtasks, including ATE, AOPE, and ASC, which can indirectly reflect the advantages of ASTE methods. Figure 11 compares our method and the backbone processing these subtasks. It suggests that the data augmentation module and the task feedback module have remarkably improved the efficiency in dealing with span extraction tasks (ATE, AOPE) and sentiment classification tasks (ASC).

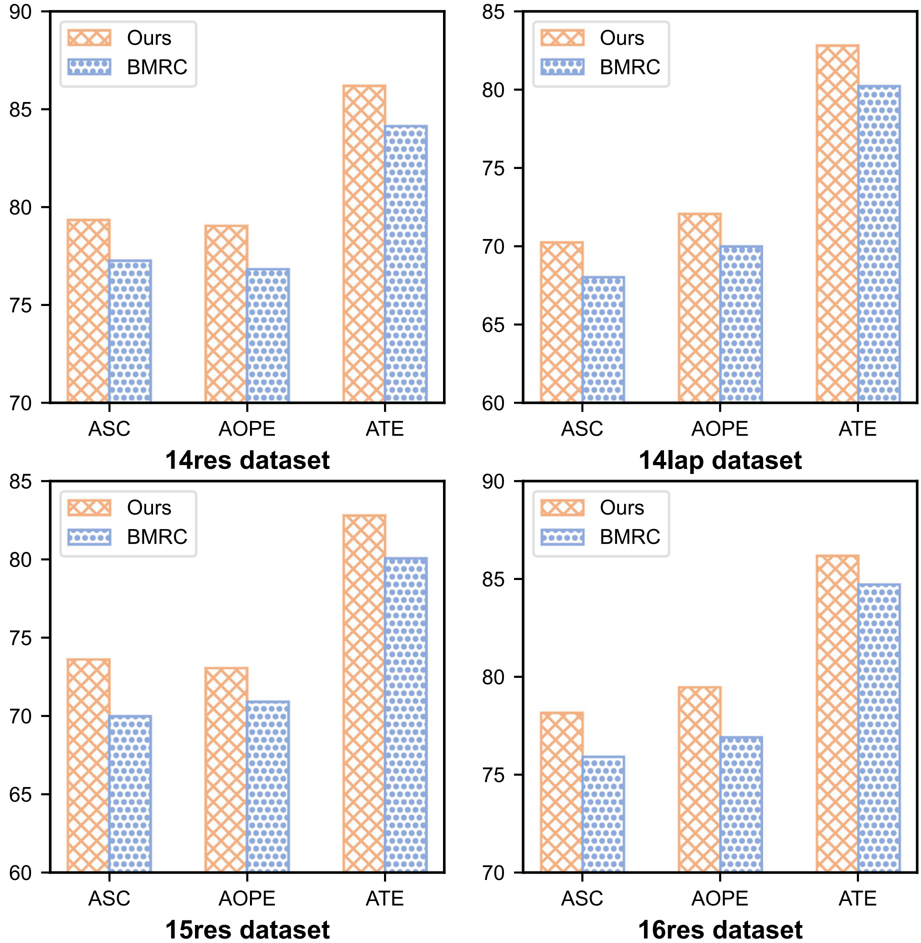

Fig. 11 Performance for ATE, AOPE, and ASC tasks in F1 score on the test set. Our method and BMRC are marked in orange and blue, respectively

5 Conclusion

In this paper, we introduce two modules on the basis of BMRC for ASTE, including data augmentation and task feedback. In order to reduce the effect of data scarcity, we provide mask prediction and mark replacement to enrich the dataset. We divide ASTE into two tasks: span extraction and sentiment classification. Our span extraction model utilizes the direct extraction method to capture spans, which avoids error accumulation and adapts to different data distributions. Our sentiment classification model establishes a task feedback module through feature fusion to tackle the separation of both tasks. The ablation studies on comparison with the BMRC verify the effectiveness of these two modules. We conduct experiments on four benchmark datasets for ASTE and our approach outperforms previous state-of-the-art methods in most cases.

Author contributions Conceptualization: Shu Liu; Methodology: Shu Liu, Tingting Lu, Kaiwen Li; Formal analysis and investigation: Tingting Lu, Kaiwen Li; Writing - original draft preparation: Tingting Lu; Writing - review and editing: Shu Liu; Funding acquisition: Shu Liu, Kaiwen Li; Resources: Weihua Liu; Supervision: Weihua Liu.

Funding This work was supported in part by Hunan Provincial Natural Science Foundation of China under Grant 2023JJ30700, and in part by the Fundamental Research Funds for the Central Universities of Central South University. We are grateful for resources from the High Performance Computing Center of Central South University.

Data availability No datasets were generated or analysed during the current study.

Code availability The code is available at https://github.com/zhenzhen313/BMRC-with-DA-and-TF.

Declarations

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

References

AlBadani, B., Dong, J., Shi, R., et al. (2023). Smgnn: span-to-span multi-channel graph neural network for aspect-sentiment triplet extraction. *Journal of Intelligent Information Systems, 61*(3), 695–715. https://doi.org/10.1007/s10844-023-00794-0

Cahuantzi, R., Chen, X., & Güttel, S. (2023). A comparison of LSTM and GRU networks for learning symbolic sequences. In K. Arai (Ed.), *Intelligent Computing* (Vol. 739, pp. 771–785). Cham: Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-37963-5_53

Chen, H., Zhai, Z., Feng, F., et al. (2022a). Enhanced multi-channel graph convolutional network for aspect sentiment triplet extraction. In: S. Muresan, P. Nakov, & A. Villavicencio (Eds.), *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics* (Volume 1: Long Papers, pp. 2974–2985). Association for Computational Linguistics, Dublin, Ireland. https://doi.org/10.18653/v1/2022.acl-long.212

Chen, S., Wang, Y., Liu, J., et al. (2021). Bidirectional machine reading comprehension for aspect sentiment triplet extraction. *Proceedings of the AAAI Conference on Artificial Intelligence, 35*(14), 12666–12674. https://doi.org/10.1609/aaai.v35i14.17500

Chen, Y., Keming, C., Sun, X., et al. (2022b). A span-level bidirectional network for aspect sentiment triplet extraction. In: Y., Goldberg, Z., Kozareva, & Y., Zhang (Eds.), *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing* (pp. 4300–4309). Association for Computational Linguistics, Abu Dhabi, United Arab Emirates. https://doi.org/10.18653/v1/2022.emnlp-main.289

Chen, Z., & Qian, T. (2020). Relation-aware collaborative learning for unified aspect-based sentiment analysis. In D. Jurafsky, J. Chai, N. Schluter, et al. (Eds.), *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 3685–3694). Association for Computational Linguistics, Online. https://doi.org/10.18653/v1/2020.acl-main.340

Dave, K., Lawrence, S., & Pennock, D. M. (2003). Mining the peanut gallery: opinion extraction and semantic classification of product reviews. *Proceedings of the 12th International Conference on World Wide Web* (pp. 519–528). Association for Computing Machinery, New York, NY, USA, WWW '03. https://doi.org/10.1145/775152.775226

Devlin, J., Chang, M. W., Lee, K., et al. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In J. Burstein, C. Doran, & T. Solorio (Eds.), *Proceedings of the 2019 Conference* *of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1, Long and Short Papers* (pp. 4171–4186). Minneapolis, Minnesota: Association for Computational Linguistics. https://doi.org/10.18653/v1/N19-1423

Di Caro, L., & Grella, M. (2013). Sentiment analysis via dependency parsing. *Computer Standards & Interfaces, 35*(5), 442–453. https://doi.org/10.1016/j.csi.2012.10.005

Fei, H., Ren, Y., Zhang, Y., et al. (2023). Nonautoregressive encoder-decoder neural framework for end-to-end aspect-based sentiment triplet extraction. *IEEE Transactions on Neural Networks and Learning Systems, 34*(9), 5544–5556. https://doi.org/10.1109/TNNLS.2021.3129483

Hu, M., & Liu, B. (2004). Mining and summarizing customer reviews. *Proceedings of the Tenth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 168–177). Association for Computing Machinery, New York, NY, USA, KDD '04. https://doi.org/10.1145/1014052.1014073

Karimi, A., Rossi, L., & Prati, A. (2021). AEDA: An easier data augmentation technique for text classification. In M. F. Moens, X. Huang, L. Specia et al. (Eds.), *Findings of the Association for Computational Linguistics: EMNLP 2021* (pp. 2748–2754). Association for Computational Linguistics, Punta Cana, Dominican Republic. https://doi.org/10.18653/v1/2021.findings-emnlp.234

Li, X., Bing, L., Li, P., et al. (2019). A unified model for opinion target extraction and target sentiment prediction. *Proceedings of the AAAI Conference on Artificial Intelligence, 33*(01), 6714–6721. https://doi.org/10.1609/aaai.v33i01.33016714

Li, Y., Cohn, T., & Baldwin, T. (2017). Robust training under linguistic adversity. In M. Lapata, P. Blunsom, & A. Koller (Eds.), *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 2, Short Papers* (pp. 21–27). Valencia, Spain: Association for Computational Linguistics. https://doi.org/10.18653/v1/e17-2004

Liang, B., Luo, W., Li, X., et al. (2021). Enhancing aspect-based sentiment analysis with supervised contrastive learning. *Proceedings of the 30th ACM International Conference on Information & Knowledge Management* (pp. 3242–3247). Association for Computing Machinery, New York, NY, USA, CIKM '21. https://doi.org/10.1145/3459637.3482096

Lin, C., & He, Y. (2009). Joint sentiment/topic model for sentiment analysis. *Proceedings of the 18th ACM Conference on Information and Knowledge Management* (pp. 375–384). Association for Computing Machinery, New York, NY, USA, CIKM '09. https://doi.org/10.1145/1645953.1646003

Liu, S., Li, K., & Li, Z. (2022). A robustly optimized BMRC for aspect sentiment triplet extraction. In M. Carpuat, M. C. de Marneffe, & I. V. Meza Ruiz (Eds.), *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* (pp. 272–278). Association for Computational Linguistics, Seattle, United States. https://doi.org/10.18653/v1/2022.naacl-main.20

Loshchilov, I., & Hutter, F. (2017). Decoupled weight decay regularization. Preprint retrieved from http://arxiv.org/abs/1711.05101. https://doi.org/10.48550/arXiv.1711.05101

Mao, Y., Shen, Y., Yu, C., et al. (2021). A joint training dual-MRC framework for aspect based sentiment analysis. *Proceedings of the AAAI Conference on Artificial Intelligence, 35*(15), 13543–13551. https://doi.org/10.1609/aaai.v35i15.17597

Peng, H., Xu, L., Bing, L., et al. (2020). Knowing what, how and why: A near complete solution for aspect-based sentiment analysis. *Proceedings of the AAAI Conference on Artificial Intelligence, 34*(05), 8600–8607. https://doi.org/10.1609/aaai.v34i05.6383

Pontiki, M., Galanis, D., Pavlopoulos, J., et al. (2014). SemEval-2014 task 4: Aspect based sentiment analysis. In P. Nakov & T. Zesch (Eds.), *Proceedings of the 8th International Workshop on Semantic Evaluation (SemEval 2014)* (pp. 27–35). Dublin, Ireland: Association for Computational Linguistics. https://doi.org/10.3115/v1/S14-2004

Pontiki, M., Galanis, D., Papageorgiou, H., et al. (2015). SemEval-2015 task 12: Aspect based sentiment analysis. In P. Nakov, T. Zesch, D. Cer, et al. (Eds.), *Proceedings of the 9th International Workshop on Semantic Evaluation (SemEval 2015)* (pp. 486–495). Association for Computational Linguistics, Denver, Colorado. https://doi.org/10.18653/v1/S15-2082

Pontiki, M., Galanis, D., Papageorgiou, H., et al. (2016). SemEval-2016 task 5: Aspect based sentiment analysis. In S. Bethard, M. Carpuat, D. Cer et al. (Eds.), *Proceedings of the 10th International Workshop on Semantic Evaluation (SemEval-2016)* (pp. 19–30). Association for Computational Linguistics, San Diego, California. https://doi.org/10.18653/v1/S16-1002

Ravi, K., & Ravi, V. (2015). A survey on opinion mining and sentiment analysis: Tasks, approaches and applications. *Knowledge-Based Systems, 89*, 14–46. https://doi.org/10.1016/j.knosys.2015.06.015

Ren, S., Zhang, J., Li, L., et al. (2021). Text AutoAugment: Learning compositional augmentation policy for text classification. In M. F. Moens, X. Huang, L. Specia, et al. (Eds.), *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing* (pp. 9029–9043). Association for

Computational Linguistics, Online and Punta Cana, Dominican Republic. https://doi.org/10.18653/v1/2021.emnlp-main.711

Singh, L. G., & Singh, S. R. (2021). Empirical study of sentiment analysis tools and techniques on societal topics. *Journal of Intelligent Information Systems, 56*(2), 379–407. https://doi.org/10.1007/s10844-020-00616-7

Sun, L., Xia, C., Yin, W., et al. (2020). Mixup-transformer: Dynamic data augmentation for NLP tasks. In D. Scott, N. Bel, & C. Zong (Eds.), *Proceedings of the 28th International Conference on Computational Linguistics* (pp. 3436–3440). International Committee on Computational Linguistics, Barcelona, Spain (Online). https://doi.org/10.18653/v1/2020.coling-main.305

Toh, Z. (2014). DLIREC: Aspect term extraction and term polarity classification system. In P. Nakov & T. Zesch (Eds.), *Proceedings of the 8th International Workshop on Semantic Evaluation (SemEval 2014)* (pp. 235–240). Dublin, Ireland: Association for Computational Linguistics.

Wang, B., Ding, L., Zhong, Q., et al. (2022). A contrastive cross-channel data augmentation framework for aspect-based sentiment analysis. In N. Calzolari, C. R. Huang, H. Kim et al. (Eds.), *Proceedings of the 29th International Conference on Computational Linguistics* (p. 6691–6704). International Committee on Computational Linguistics, Gyeongju, Republic of Korea. https://aclanthology.org/2022.coling-1.581

Wang, Q., Wen, Z., Zhao, Q., et al. (2021). Progressive self-training with discriminator for aspect term extraction. In M. F., Moens, X. Huang, L. Specia et al. (Eds.), *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing* (p. 257–268). Association for Computational Linguistics, Online and Punta Cana, Dominican Republic. https://doi.org/10.18653/v1/2021.emnlp-main.23

Wang, Y., Chen, Z., & Chen, S. (2023). Es-aste: enhanced span-level framework for aspect sentiment triplet extraction. *Journal of Intelligent Information Systems, 60*(3), 593–612. https://doi.org/10.1007/s10844-023-00783-3

Wei, J., & Zou, K. (2019). EDA: Easy data augmentation techniques for boosting performance on text classification tasks. In K. Inui, J. Jiang, V. Ng et al. (Eds.), *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)* (pp. 6382–6388). Association for Computational Linguistics, Hong Kong, China. https://doi.org/10.18653/v1/D19-1670

Wu, Z., Ying, C., Zhao, F., et al. (2020). Grid tagging scheme for aspect-oriented fine-grained opinion extraction. In T. Cohn, Y. He, & Y. Liu (Eds.), *Findings of the Association for Computational Linguistics: EMNLP 2020* (pp. 2576–2585). Association for Computational Linguistics, Online. https://doi.org/10.18653/v1/2020.findings-emnlp.234

Xu, L., Li, H., Lu, W., et al. (2020). Position-aware tagging for aspect sentiment triplet extraction. In B. Webber, T. Cohn, Y. He et al. (Eds.), *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (pp. 2339–2349). Association for Computational Linguistics, Online. https://doi.org/10.18653/v1/2020.emnlp-main.183

Xu, L., Chia, Y. K., & Bing, L. (2021). Learning span-level interactions for aspect sentiment triplet extraction. In C. Zong, F. Xia, W. Li et al. (Eds.), *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)* (pp. 4755–4766). Association for Computational Linguistics, Online. https://doi.org/10.18653/v1/2021.acl-long.367

Yan, H., Dai, J., Ji, T., et al. (2021). A unified generative framework for aspect-based sentiment analysis. In C. Zong, F. Xia, W. Li et al. (Eds.), *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)* (pp. 2416–2429). Association for Computational Linguistics, Online. https://doi.org/10.18653/v1/2021.acl-long.188

Yu, G., Liu, L., Jiang, H., et al. (2023). Making better use of training corpus: Retrieval-based aspect sentiment triplet extraction via label interpolation. In A. Rogers, J. Boyd-Graber, & N. Okazaki (Eds.), *Findings of the Association for Computational Linguistics: ACL 2023* (pp. 4914–4927). Association for Computational Linguistics, Toronto, Canada. https://doi.org/10.18653/v1/2023.findings-acl.303

Zhang, C., Li, Q., Song, D., et al. (2020). A multi-task learning framework for opinion triplet extraction. In T. Cohn, Y. He, & Y. Liu (Eds.), *Findings of the Association for Computational Linguistics: EMNLP 2020* (pp. 819–828). Association for Computational Linguistics, Online. https://doi.org/10.18653/v1/2020.findings-emnlp.72

Zhang, L., & Liu, B. (2014). *Aspect and entity extraction for opinion mining* (pp. 1–40). Berlin, Heidelberg: Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-40837-3_1

Zhang, W., Li, X., Deng, Y., et al. (2021). Towards generative aspect-based sentiment analysis. In C. Zong, F. Xia, W. Li et al. (Eds.), *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 2: Short*

Papers) (pp. 504–510). Association for Computational Linguistics, Online. https://doi.org/10.18653/v1/2021.acl-short.64

Zhang, Y., Ding, Q., Zhu, Z., et al. (2022). Enhancing aspect and opinion terms semantic relation for aspect sentiment triplet extraction. *Journal of Intelligent Information Systems, 59*(2), 523–542. https://doi.org/10.1007/s10844-022-00710-y

Zhou, S., & Qian, T. (2023). On the strength of sequence labeling and generative models for aspect sentiment triplet extraction. In A. Rogers, J. Boyd-Graber, & N. Okazaki (Eds.), *Findings of the Association for Computational Linguistics: ACL 2023* (pp. 12038–12050). Toronto, Canada: Association for Computational Linguistics. https://doi.org/10.18653/v1/2023.findings-acl.762

Zhu, L., Zhu, X., Guo, J., et al. (2023). Exploring rich structure information for aspect-based sentiment classification. *Journal of Intelligent Information Systems, 60*(1), 97–117. https://doi.org/10.1007/s10844-022-00729-1